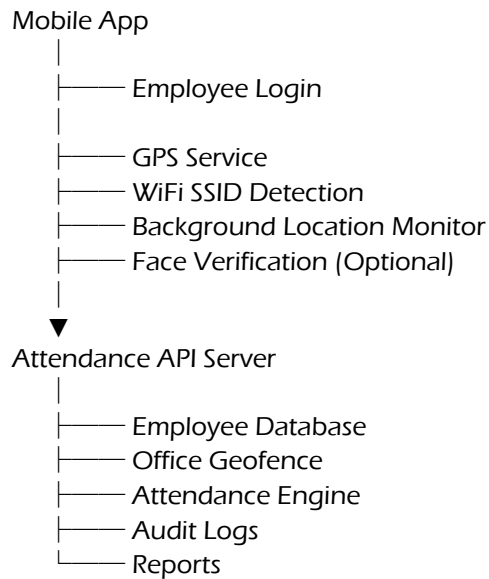


A robust employee attendance system should not rely on a single method such as GPS or Wi-Fi because each has weaknesses. Instead, use a **multi-factor location verification model** that combines SSID, GPS, geofencing, and periodic validation.

System Architecture



Clock-In Logic

Step 1 – Employee Login

- Authenticate using Employee ID/Password
- Optional Biometric Authentication (Fingerprint/Face)

Step 2 – Verify Office Location

Option A – WiFi SSID

Current SSID == HOTEL-STAFF

If YES

Confidence = High

If NO

Proceed to GPS verification.

Option B – GPS

Obtain

Latitude
Longitude
Accuracy

Calculate distance from office.

Example

Office Coordinates

25.253174
55.365673

Allowed Radius

100 meters

If

Distance $\leq$ 100m

Clock In Allowed

Else

Reject

Step 3 – Record Clock In

Store

Employee ID
Date
Time
Latitude
Longitude
SSID
Phone ID
Battery Level

Network Type
Accuracy

During Working Hours

Instead of tracking every minute (which drains battery), verify every **15–30 minutes**.

Example

09:00 Clock In

09:30 Verify

10:00 Verify

10:30 Verify

11:00 Verify

...

Each verification records

Latitude

Longitude

SSID

Timestamp

Accuracy

Verification Logic

SSID Available?

YES

↓

Office SSID?

YES

↓

Status = Present

Otherwise

GPS within office?

YES

↓

Status = Present

Otherwise

Outside Office

Consecutive Failures

Example

09:30 Outside

09:45 Outside

10:00 Outside

System

Send Notification

"You appear to have left the office.
Please return or provide justification."

Supervisor receives alert.

Auto Clock-Out

At scheduled shift end

Example

6:00 PM

Verify

SSID

or

GPS

If inside office

Clock Out

If outside

Use last verified location.

Manual Clock-Out

Employee presses

Clock Out

System verifies

SSID OR GPS

If verified

Attendance Closed

Else

Require reason

Official Visit

Work From Home

Customer Visit

Manager Approval

Preventing Fraud

GPS Spoofing

Detect

Mock Location Enabled

Developer Options

Fake GPS Apps

Reject attendance.

WiFi Validation

Don't only compare SSID.

Also validate

SSID

BSSID (Router MAC)

Signal Strength

Example

SSID

DIH-STAFF

BSSID

00:12:34:56:78:90

This prevents users from creating a fake hotspot with the same SSID.

Device Binding

Register

Employee

↓

Mobile IMEI (Android)

or

Device UUID (iPhone)

Only approved devices can record attendance.

Face Verification (Optional)

During Clock In

Take Selfie

↓

AI Face Match

↓

Attendance Recorded

This prevents buddy punching.

Attendance Status Rules

Condition	Status
SSID Match	Present
GPS داخل Geofence	Present
GPS Outside	Outside Office
No Internet	Store Offline
GPS Disabled	Prompt User
Fake GPS	Reject
SSID Missing + GPS Outside	Reject

Offline Mode

If Internet unavailable

Store locally

Clock In

Location

SSID

Timestamp

Synchronize automatically when connectivity is restored.

Background Monitoring

The app should use **low-power background location services**:

- Trigger checks every **15–30 minutes** rather than continuous tracking.
 - Immediately verify when the device reconnects to the office Wi-Fi or re-enters the geofence.
 - Pause monitoring after clock-out to preserve battery.
-

Suggested Database Tables

Employees

EmployeeID

Name

Department

RegisteredDevice

Status

Attendance

AttendancID

EmployeeID

ClockIn

ClockOut

WorkHours

Status

Location Logs

AttendancID

Time

Latitude

Longitude

SSID

BSSID

Accuracy

InsideOffice (Yes/No)

Recommended Attendance Workflow

1. Employee logs in using biometric authentication.
2. Verify **office Wi-Fi SSID + BSSID**. If matched, accept immediately.
3. If Wi-Fi is unavailable, verify the employee is within a **100-meter GPS geofence**.
4. Capture clock-in time, location, and device information.
5. Perform **background verification every 15–30 minutes** using Wi-Fi first and GPS as a fallback.
6. Generate alerts after multiple consecutive failed location checks.
7. Verify location again during clock-out.
8. Synchronize any offline records when connectivity is restored.
9. Provide managers with attendance summaries, location history, exceptions, and audit logs.

This hybrid approach provides high accuracy while minimizing battery consumption and reducing opportunities for attendance fraud.